

Notes: Jegadeesh, Kim, Krische & Lee (JF, 2004)
Analyzing the Analysts: When Do Recommendations Add Value?

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1 Big picture

- The paper asks whether sell-side analyst recommendations add investment value after controlling for known cross-sectional return predictors.
- It separates consensus recommendation levels (CON) from changes in consensus recommendations (CHGCON).
- Analysts generally recommend glamour stocks: momentum winners, high-growth, high-volume, and relatively expensive firms.
- Static recommendation levels add value only conditionally and can be harmful when quantitative signals are unfavorable.
- Recommendation changes are more robust return predictors and contain information orthogonal to many quantitative signals.

2 Incremental contribution

- Connects analyst recommendation research to the asset-pricing anomaly literature.
- Benchmarks analyst recommendations against 12 known return predictors rather than testing recommendations in isolation.
- Shows that recommendation levels and recommendation changes have distinct economic content.
- Documents analyst glamour bias and explains why static recommendations can fail.
- Shows that analyst changes are more incrementally useful than recommendation levels.
- Provides a practical framework for combining human analyst signals with quantitative screens.

3 Data and measurement

The sample comes from Zacks analyst recommendations from 1985 through 1998, matched to CRSP, I/B/E/S, and quarterly COMPUSTAT. Recommendations are reverse-coded so that higher values indicate more favorable ratings. For each firm-quarter:

$$CON_{it} = \text{mean current outstanding recommendations}, \quad CHGCON_{it} = CON_{it} - CON_{i,t-1}.$$

The paper constructs 12 quantitative signals: RETP, RET2P, FREV, SUE, TURN, EP, BP, LTG, SG, TA, CAPEX, and SIZE. These are combined into Momentum, Contrarian, and QScore summary variables.

Detailed interpretation. The key research design is to ask whether analyst signals have value beyond these quantitative predictors.

4 Empirical specifications

$$\begin{aligned} RET_{iq}^F &= \alpha_q + \beta_q CON_{iq} + \gamma_q QuantSignals_{iq} + \epsilon_{iq}, \\ RET_{iq}^F &= \alpha_q + \beta_q CHGCON_{iq} + \gamma_q QuantSignals_{iq} + \epsilon_{iq}. \end{aligned}$$

The paper also uses Spearman rank correlations, portfolio sorts, two-way sorts by QScore and analyst signal, and cumulative excess-return tests.

5 Identification and empirical design

The paper is not causal. It is an incremental-predictability design. The identification problem is that analysts recommend stocks with characteristics that already predict returns. The solution is to condition on the 12 quantitative predictors and ask what remains. The core finding is that CON is largely explained by characteristics, whereas CHGCON retains residual predictive power.

6 Tables and figures

6.1 Figure 1: Data accumulation periods relative to portfolio formation date

Read: Figure 1 defines the timing for recommendation levels, recommendation changes, investment signals, and future returns.

Detailed interpretation. It prevents look-ahead bias by ensuring all predictors are measured before the future-return window.

Economic message. Timing discipline is essential for a credible predictability test.

asures. For a consensus recommendation level observed at the end of quarter t , we use market-related data (past returns and trading volume) and analyst-related data that are collected up to 12 months prior to the end of quarter t . For accounting-related data, we identify q as the most recent quarter for which an

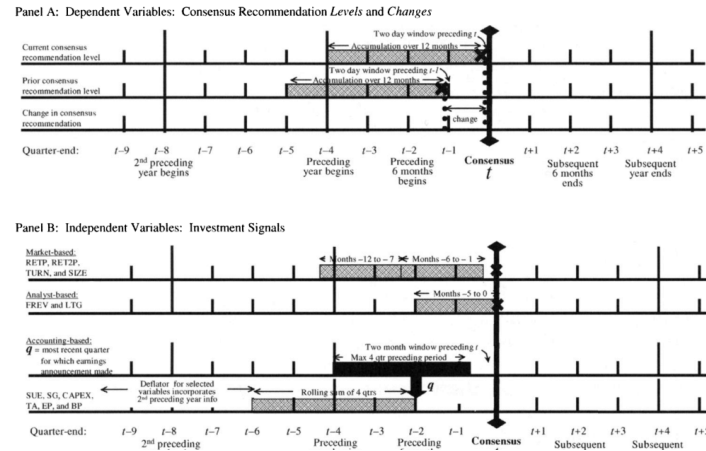


Figure 1

6.2 Tables I–II: Sample and recommendations

Read: Table I describes sample firms; Table II describes analyst recommendation levels and changes. Analysts are generally optimistic and recommendation levels mean revert.

Detailed interpretation. These tables show that CON and CHGCON are distinct signals.

Economic message. Static optimism is different from new analyst information.

Table I
Description of Sample Firms

This table provides descriptive statistics on the firms included in our sample. Our sample consists of all firms with current individual stock recommendations in the Zacks database (defined as recommendations that have been outstanding for less than one year), provided the firm also has the required CRSP, COMPUSTAT, and I/B/E/S information. Panel A reports the mean number of observations per quarter for each individual year and the average level of consensus recommendation. We reverse-score the recommendations from five (strong buy) to one (strong sell). The consensus recommendation is defined as the mean of all individual recommendations two days prior to the end of each calendar quarter. Mean Consensus in the table is the average of the consensus recommendations at the end of each quarter. Panels B and C report the sample distributions across exchanges and size deciles. We include firms with CRSP codes for regional exchanges and the OTC market (0.6% of observations) with the Nasdaq sample. We obtain the exchange listing as of the time of the consensus recommendation. The sample period is from 1985:1 through 1998:4.

Panel A: Sample Distribution across Years			
Year	Observations Per Quarter	% of total	Mean Consensus
1985	404.75	3.0	3.21
1986	618.75	4.5	3.45
1987	670.75	4.9	3.61
1988	714.50	5.3	3.65
1989	854.25	6.3	3.56
1990	946.75	7.0	3.60
1991	966.50	7.1	3.58
1992	1,009.25	7.4	3.68
1993	1,137.25	8.4	3.70
1994	1,291.75	9.5	3.84
1995	1,201.00	8.8	3.82
1996	1,243.00	9.1	3.79
1997	1,257.00	9.2	3.92
1998	1,284.50	9.4	3.97
Sample Average	971.43		3.67

Panel B: Sample Distribution across Exchanges		
Exchange	Mean Observations Per Quarter	Mean % of Sample Per Quarter
Nasdaq	575.82	57.0
NYSE	263.96	28.7
AMEX	131.64	14.3
Mean Quarterly Sample	971.43	100.0%

(a) Table I, part 1

Table II
Description of Analyst Recommendations

This table provides descriptive statistics on the analyst recommendations in our sample. We use all individual recommendations in the Zacks database that have been outstanding for less than one year. We reverse-score the recommendations from five (strong buy) to one (strong sell). We define consensus recommendation as the mean of all individual recommendations two days prior to the end of each calendar quarter. Firms are sorted into quintiles at the beginning of the next quarter based on either the level of the consensus recommendation, or the change in the consensus recommendation over the most recent quarter. Firm-quarter is the unit of observation in Panels A and B. We compute the variables in the table each quarter, and report the summary statistics of the time-series. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. Panel A reports summary statistics for each of the level quintiles. The column "Coded as" reports the numerical values assigned to each quintile. The column "Mean Obs" is the average number of stocks in the sample per quarter. Panel B reports the summary statistics for each of the change quintiles. The middle quintile contains all observations with no change in consensus recommendations. Panel C reports the frequency distribution of observations in each change quintile, conditional on its level quintile membership in the prior quarter.

Quintile	Coded as	Mean Obs	Mean	Std Dev	Minimum	Maximum
Panel A: Consensus Recommendation Level Quintiles (Strong BUY = 5, HOLD = 3, Strong SELL = 1)						
Best = BUY	1.00	176.91	4.62	0.140	4.42	4.87
	0.75	194.77	4.07	0.171	3.77	4.38
	0.50	200.27	3.72	0.196	3.27	4.04
	0.25	186.32	3.37	0.179	2.95	3.75
Worst = SELL	0.00	213.16	2.76	0.238	1.81	3.13
Panel B: Consensus Recommendation Change Quintiles						
Best = Increase	1.00	192.07	0.52	0.121	0.37	1.06
	0.75	144.75	0.12	0.050	0.05	0.35
	0.50	294.24	0.00	0.003	-0.01	0.02
	0.25	144.80	-0.11	0.032	-0.23	-0.06
Worst = Decrease	0.00	198.60	-0.55	0.085	-0.88	-0.41
Panel C: Percentage of Sample in Recommendation Change Quintile, Conditioned on Prior Consensus Level Quintile						
Prior Consensus Quintile	Recommendation Change Quintile					Total Sample
	Worst = Decrease			Best = Increase		
Best = BUY	32.2%	14.8%	37.3%	9.8%	5.8%	18.3%
	25.5%	17.6%	25.2%	16.7%	15.1%	20.2%
	20.1%	19.3%	21.5%	18.2%	20.9%	20.7%
	16.7%	17.4%	19.6%	20.3%	26.0%	19.1%
Worst = SELL	9.3%	5.9%	46.4%	9.5%	29.0%	21.8%
Total Sample	20.4%	14.9%	30.2%	14.9%	19.7%	100.0%

(b) Table II

6.3 Tables III–V: Basic predictive power

Read: Table III confirms that analyst recommendations predict future returns. Tables IV and V show that quantitative signals and QScore also strongly predict returns.

Detailed interpretation. These tables create the benchmark for the incremental-value tests.

Economic message. Analysts must beat a strong quantitative benchmark, not a zero-information benchmark.

Table III
Analyst Recommendations and Future Returns

This table examines the correlation between analyst recommendations and future returns. Future returns are defined as the market-adjusted return in the six months after the month of the recommendation (RETF). Two different measures of analyst recommendations are used: (1) the consensus recommendation level (CON) and (2) the change in the consensus measured over the prior quarter (CHGCON). Panel A reports the Spearman rank correlation between each analyst recommendation measure and future returns. The column "Continuous Expl. Variable" uses the actual values of recommendation levels and changes, and the column "Categorical Expl. Variable" uses the quintile scores for these variables from Table II. Panel B reports future returns for firms grouped by their consensus recommendation levels (CON), and Panel C reports future returns grouped by the change in the consensus recommendation (CHGCON). We compute the statistics in the table each quarter, and report the means and medians of the time-series. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively. The significance level for the Median is based on the Wilcoxon signed-rank test, and for the Mean is based on *t*-statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

Panel A: Spearman Rank Correlations with Future Returns				
Explanatory Variable	Continuous Expl. Variable		Categorical Expl. Variable	
	Mean	Median	Mean	Median
Consensus Level ("CON")	+0.0312**	+0.0276**	+0.0311**	+0.0350**
Consensus Change ("CHGCON")	+0.0333***	+0.0384***	+0.0317***	+0.0286***

Panel B: Market-adjusted Returns by Consensus Recommendation Level Quintile			
Quintile	Coded as	Mean	Median
Best = BUY	1.00	−0.003	−0.024
	0.75	−0.008	−0.024
	0.50	−0.015	−0.032
	0.25	−0.018	−0.033
Worst = SELL	0.00	−0.027	−0.055
BUY − SELL		+0.023**	+0.034***

Panel C: Market-adjusted Returns by Consensus Recommendation Change Quintile			
Quintile	Coded as	Mean	Median
Best = Increase	1.00	−0.004	−0.025
	0.75	−0.007	−0.015
	0.50	−0.022	−0.044
	0.25	−0.004	−0.023
Worst = Decrease	0.00	−0.031	−0.051
Increase − Decrease		+0.027***	+0.031***

Table III

Table IV
Quantitative Investment Signals and Future Returns

This table presents the Spearman rank correlation between future returns (RET_F) and various quantitative investment signals. We define RET_F as the market-adjusted return in the six months following the month of the recommendation. The 12 quantitative investment signals are described in detail in Appendix A. For variables that are positively (negatively) correlated with future returns, the binary variable assumes a value of one if the explanatory variable is higher (lower) than the median for that quarter, and zero otherwise. The net portfolio return is the mean difference in future returns between the portfolio of firms with the binary variable equal to one, and the portfolio of firms with the binary variable equal to zero. Our estimates are formed once each quarter for the 56 quarters from 1985:1 to 1998:4. We aggregate these quarterly estimates and report their time-series averages. The column "% Positive Returns" presents the percentage of the 56 quarters in which the net portfolio return was greater than 0%. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively, based on *t*-statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

Explanatory Variable	Continuous Explanatory Variable Correlation	Binary Explanatory Variable			
		Normative Definition	Correlation	Mean Net Portfolio Return	% Positive Returns
RET _P	+0.080***	1 if greater than median, 0 otherwise	+0.064***	+0.032***	76.79
RET _{2P}	+0.043***	1 if greater than median, 0 otherwise	+0.039***	+0.013*	62.50
TURN	-0.034**	1 if less than median, 0 otherwise	+0.033**	+0.002	62.50
SIZE	+0.088***	1 if less than median, 0 otherwise	-0.077***	-0.016	35.71
FREV	+0.099***	1 if greater than median, 0 otherwise	+0.091***	+0.042***	83.93
LTG	-0.006	1 if less than median, 0 otherwise	+0.008	-0.000	53.57
SUE	+0.053***	1 if greater than median, 0 otherwise	+0.040***	+0.018**	67.86
SG	-0.025*	1 if less than median, 0 otherwise	+0.025*	+0.004	57.14
TA	-0.081***	1 if less than median, 0 otherwise	+0.063***	+0.029***	85.71
CAPEX	-0.021*	1 if less than median, 0 otherwise	+0.023**	+0.015***	69.64
BP	-0.016	1 if greater than median, 0 otherwise	-0.010	-0.000	50.00
EP	+0.038**	1 if greater than median, 0 otherwise	+0.029**	+0.004	55.36

(a) Table IV

Table V
Summary Quantitative Variables and Future Returns

This table examines the correlation between the three summary quantitative variables and future returns. The dependent variable is future returns (RET_F), defined as the market-adjusted return in the six months after portfolio formation. We use three different summary variables: Momentum (the sum of four momentum signals: RET_P, RET_{2P}, FREV, SUE), Contrarian (the sum of seven contrarian signals: EP, BP, TURN, LTG, SG, TA, CAPEX), and QScore (the sum of all 12 investment signals, including SIZE). Appendix A describes the signals. We combine extreme categories of the summary variables, such that the extreme groups have an average of at least about 100 stocks per quarter. Panel A reports the Spearman rank correlation between each sum measure and future returns. Panels B, C, and D report future returns for firms grouped by QScore categories, Momentum categories, and Contrarian categories, respectively. We compute the variables in the table each quarter, and report the means and medians of the time-series. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively. The significance level for the Median is based on the Wilcoxon signed-rank test, and that for the Mean is based on *t*-statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

Panel A: Spearman Rank Correlations with Future Returns					
Explanatory Variable	Mean		Median		
QScore	0.0830***		0.0850***		
Momentum	0.0865***		0.0907***		
Contrarian	0.0488***		0.0539***		

Panel B: Market-adjusted Returns by QScore Category					
	FScore Sum	Coded as	Mean Obs Per Qtr	Mean	Median
Best =	9, 10, 11, 12	1.00	103.96	+0.0194	-0.0060
	8	0.83	116.36	-0.0025	-0.0236
	7	0.67	162.71	-0.0080	-0.0284
	6	0.50	181.16	-0.0097	-0.0356
	5	0.33	171.36	-0.0263	-0.0420
Worst =	4	0.17	118.93	-0.0334	-0.0566
	0, 1, 2, 3	0.00	99.46	-0.0506	-0.0760
Best - Worst				+0.0699***	+0.0685***

Panel C: Market-adjusted Returns by Momentum Category					
	Momentum Sum	Coded as	Mean Obs Per Qtr	Mean	Median
Best =	4	1.00	161.84	+0.0141	-0.0047
	3	0.75	217.88	-0.0012	-0.0219
	2	0.50	195.77	-0.0165	-0.0308
	1	0.25	217.89	-0.0305	-0.0548
Worst =	0	0.00	160.57	-0.0432	-0.0624
Best - Worst				+0.0573***	+0.0620***

Panel D: Market-adjusted Returns by Contrarian Category					
	Contrarian Sum	Coded as	Mean Obs per Qtr	Mean	Median
Best =	6, 7	1.00	113.50	+0.0032	-0.0200
	5	0.80	173.45	-0.0094	-0.0318
	4	0.60	206.77	-0.0176	-0.0426
	3	0.40	188.07	-0.0194	-0.0451
	2	0.20	145.05	-0.0161	-0.0417

(b) Table V

6.4 Tables VI–VIII: Analyst bias and multivariate tests

Read: Table VI shows analysts favor glamour stocks. Table VII shows recommendations are explained by quantitative characteristics. Table VIII shows QCON loses value after controls, while QCHGCON remains useful.

Detailed interpretation. The level of analyst recommendations is style-biased; changes are more informational.

Economic message. The main investment lesson is to treat levels and changes differently.

Table VI
Descriptive Statistics by Consensus Recommendation Quintile
This table examines the relation between the level of the consensus recommendation and 12 investment signals. The signals are described in detail in Appendix A. Our unit of observation is firm-quarter. To construct Panel A, we sort all firms by their consensus recommendation levels and group into quintiles each quarter. Table values are the time series means of the investment signal for each recommendation quintile. "Normative Direction" indicates the sign of the variable's correlation with future returns from prior studies. Correlation is the mean Spearman rank correlation on the consensus recommendation and a given investment signal across the quarters. "Actual Direction" indicates the sign of the relation when an correlation is 10% or higher. When the absolute value of correlation is less than 10%, the table indicates the actual direction as "?". Panel B presents this analysis for changes in consensus recommendations. The sample period is from 1985:1 through 1998:4 for recommendation levels and 1985:2 through 1998:4 for recommendation changes.

Panel A: Consensus Recommendation Level								
Investment Signal	Normative Direction	Consensus Recommendation Quintile					Correlation	Actual Direction
		BUY 1.00	0.75	0.50	0.25	SELL 0.00		
	—	5.5629	6.1999	6.4940	6.2727	5.2186	4.29%	?
Momentum Variables (Price or Earning)								
P	+	0.1508	0.1192	0.0827	0.0277	-0.0241	26.89%	+
IP	+	0.1758	0.1384	0.0946	0.0430	-0.0146	27.90%	+
I	+	-0.3274	-0.4703	-0.7619	-1.4352	-2.6510	34.59%	+
	+	1.0068	0.8898	0.5319	0.1230	-0.2711	32.10%	+
Contrarian Variables (Fundamental or Growth)								
	+	0.0550	0.0551	0.0543	0.0465	0.0362	11.89%	+
	+	0.4727	0.4832	0.5281	0.5996	0.7499	-30.11%	-
N	-	52.2900	53.1011	52.5706	50.0011	41.1517	11.82%	+
	-	1.2203	1.1875	1.1356	1.1032	1.0728	29.64%	+
	-	24.0312	20.1197	14.4616	9.7340	3.4313	27.24%	+
	-	0.0213	0.0148	0.0052	0.0018	0.0025	10.62%	+
IX	-	0.0887	0.0901	0.0897	0.0872	0.0766	4.24%	?

(a) Table VI, part 1

Table VI—Continued								
Panel B: Consensus Recommendation Change								
Investment Signal	Normative Direction	Consensus Recommendation Quintile					Correlation	Actual Direction
		BUY 1.00	0.75	0.50	0.25	SELL 0.00		
	—	5.8730	6.9277	5.0580	6.9500	5.8029	0.96%	?
Momentum Variables (Price or Earning)								
P	+	0.1111	0.0994	0.0576	0.0665	0.0137	14.58%	+
IP	+	0.0706	0.0979	0.0705	0.0954	0.0967	-3.24%	?
I	+	-0.9649	-0.7163	-1.3829	-0.9096	-1.5472	9.73%	?
	+	0.3958	0.6120	0.3033	0.6220	0.3596	1.42%	?
Contrarian Variables (Fundamental or Growth)								
	+	0.0428	0.0484	0.0440	0.0500	0.0511	-3.93%	?
	+	0.5819	0.4985	0.6440	0.4896	0.5554	3.00%	?
	-	40.5835	56.8875	40.2862	56.8940	52.1123	-3.14%	?
	-	1.1309	1.1444	1.1303	1.1500	1.1600	-5.16%	?
	-	12.6934	15.3931	10.9226	16.4032	17.3849	-5.76%	?
	-	0.0052	0.0038	0.0106	0.0058	0.0161	-5.30%	?
IX	-	0.0848	0.0908	0.0786	0.0924	0.0895	-2.25%	?

(b) Table VI, part 2

respectively. With a few exceptions, Table VII confirms the univariate results reported in Table VI. Panel A shows that the level of the consensus recommendation is generally consistent with the Momentum variables, but runs counter to the Contrarian variables. Panel B shows that changes in the consensus recommendation are also consistent with the Momentum variables, but its correlation with the Contrarian variables is less clear.

The general picture that emerges from this analysis is that analysts favorably recommend stocks with strong past operating performance and stocks that are expected to deliver healthy improvements in operating performance in the future. High SUE for the most favorably recommended stocks indicates that

Table VII
Regression of Recommendations on Explanatory Variables

This table reports the result when analyst recommendations levels and changes are regressed on various explanatory variables. Panel A (Panel B) reports results when the dependent variable is the level of (changes in) the consensus recommendation. The explanatory variables are explained in detail in Appendix A. For ease of interpretation, we standardize the explanatory variables by subtracting their means and dividing by their standard deviations each quarter. We obtain the regression estimates each quarter, and report the time-series averages of the estimates. The column "Normative Direction" indicates the sign of the variable's correlation with future returns from prior studies. The column "Actual Direction" indicates the sign of the relation when the rank correlation is 10% or higher. When the absolute value of correlation is less than 10%, the table indicates the actual direction as "?". Mean R^2 and Mean F -statistic are the averages of the respective statistics from the quarterly cross-sectional regressions. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively, based on t -statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

Panel A: Consensus Recommendation Level ("CON")				
Variable	Normative Direction	Actual Direction	Regression Coefficient	t -statistic
Intercept			+3.670	+71.29***
SIZE	—	—	-0.070	-3.72***
Momentum Variables (Price or Earning)				
RET _P	+	+	+0.101	+15.16***
RET _{2P}	+	+	+0.074	+10.08***
FREV	+	+	+0.077	+11.45***
SUE	+	+	+0.061	+7.90***
Contrarian Variables (Fundamental or Growth)				
EP	+	+	+0.041	+6.57***
BP	+	-	-0.097	-17.63***

(a) Table VII

Table VIII
Future Returns, Analyst Recommendations, and Investment Signals
This table reports regressions of future returns on analyst recommendations and on various investment signals. Future returns are defined as the t -adjusted return in the six months after the month of the recommendation (RET_F). Analyst recommendations are the quintile of the consensus recommendation level (QCON), and the quintile of the change in the consensus measured over the prior quarter (QCHGCON). Two different summary variables are used: Momentum (the sum of four momentum signals: RET_P, RET_{2P}, FREV, SUE) and Contrarian (the sum of seven contrarian signals: IP, TURN, LTG, SG, TA, CAPEX), as described in Table V. The variables QFIT_{QCON} and QFIT_{QCHGCON} are fitted values of QCON and QCHGCON. Panels A and B of Table VII, respectively. We obtain the regression estimates each quarter, and report the time-series averages of the estimates. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively, based on t -statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

Panel A: Consensus Recommendation Level Quintiles (QCON)												
	Model A1		Model A2		Model A3		Model A4		Model A5		Model A6	
	Analysts Alone		Analysts & Both Summary Variables		Analysts & Momentum Summary Variable		Analysts & Contrarian Summary Variable		Analysts & Binary Investment Signals		Analysts & Fitted Value	
netcr	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat	Estimate	t-stat
sept	-0.0257	-2.52**	-0.0702	-4.86***	-0.0440	-3.34***	-0.0467	-3.43***	-0.0684	-4.90***	-0.0336	-2.72***
return	+0.0226	2.03**	+0.0088	1.07	+0.0009	0.10	+0.0310	2.95**	+0.0076	1.05	+0.0074	1.04
return			+0.0619	5.27***	+0.0570	4.77***						
return			+0.0396	2.72***			+0.0336	2.29**				
									-0.0078	-0.93		
									+0.0175	3.45***		
									+0.0019	0.36		
									+0.0324	8.63***		
									+0.0017	0.40		
									+0.0021	0.35		
									+0.0089	1.60		
									+0.0011	0.15		
									+0.0007	0.13		
									+0.0007	0.14		
									+0.0268	8.43***		
IX									+0.0134	3.05***		
CON											+0.0300	2.32**

(b) Table VIII

6.5 Table IX, Table X, and Figure 2: Conditional and cumulative tests

Read: Table IX shows recommendation levels help mainly when QScore is favorable. Table X and Figure 2 show cumulative hedge returns.

Detailed interpretation. Recommendations and quantitative signals are complements, not substitutes.

Economic message. Use analyst changes and levels conditional on quantitative screens.

Table X

Cumulative Returns Difference over Various Holding Periods

This table reports the return differences over various holding periods following portfolio formation. We group firms by their quantitative measure (QScore, Momentum, Contrarian) and consensus recommendations. Panel A reports the mean difference in returns between the extreme consensus recommendation level quintiles (BUY-SELL) and changes quintiles (INCREASE-DECREASE) within each of the QScore categories. Panels B and C repeat the analyses for Momentum and Contrarian categories, respectively. We compute the return differences quarterly, and report the time-series means. The sample period is from 1985:1 through 1998:4 for recommendation levels and from 1985:2 through 1998:4 for recommendation changes. We indicate two-sided statistical significance at 1%, 5%, and 10% as ***, **, and * respectively, based on *t*-statistics calculated with autocorrelation consistent standard errors (as described in Appendix B).

[illegible]

(b) Table X

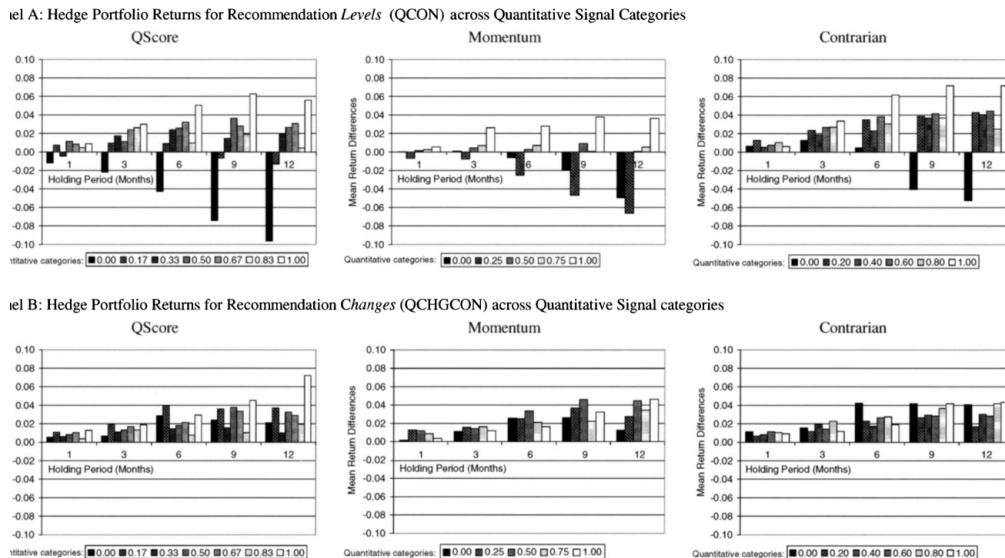


Figure 2

7 Conclusion

7.1 Core conclusion

Analyst recommendations add value conditionally. Static levels are contaminated by glamour bias, while changes are more robust predictors.

7.2 Economic intuition

Analysts tend to favor exciting, high-growth, high-volume firms. This preference sometimes conflicts with quantitative signals that forecast future returns. Recommendation changes are less tied to static style and more likely to reflect new information.

7.3 Incremental contribution

The paper shows that analyst value must be evaluated relative to known return predictors. It also demonstrates that recommendation levels and changes should be analyzed separately.

7.4 Takeaway

The best use of analyst research is not naive adherence to buy recommendations, but integration of analyst revisions with disciplined quantitative signals.